

Using Live Tags for Summarizing Surgical Videos Collaboratively

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ABSTRACT

Surgical videos serve various purposes, including training and dissemination of science. However, their production remains limited given the time needed and the limited support from commercial video editing tools. This work explores additive summarization, a novel approach to surgical video editing based on concatenating tagged segments, rather than subtracting unimportant sequences. We conducted three workshops where pairs of surgeons (expert–novice) summarized surgical videos using a custom-built prototype. The results show that tags support additive summarization, ease the creation of multiple summaries, and support collaborative work. Moreover, we observe that the expertise of residents impacts the division of work. These results show the potential of the additive approach, and open research avenues for future work studying its effectiveness.

CCS CONCEPTS

• **Human-centered computing** → **Collaborative and social computing**; **Interaction design process and methods**.

KEYWORDS

surgery, video summarization, endoscopic video, collaborative video editing, video tags

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1 INTRODUCTION

Videos play a valuable role in the field of surgery, serving various purposes such as training, discussing cases, and disseminating scientific knowledge [1, 5, 13]. They are particularly relevant because many surgical procedures are already video-assisted, for example in laparoscopic or endonasal surgery where the surgeon operates using optics connected to a camera. Thus, capturing raw surgical videos is straightforward [30]. However, for videos to be useful, they must be summarized, as they can span several hours and include repetitive gestures (sutures or dissection) as well as unusable footage (blurry or overexposed images) [11]. Empirical studies have investigated current practices on summarizing surgical videos, showing that surgeons use commercial software (e.g., iMovie) to carry out the editing process themselves, in collaboration with other surgeons [5]. However, this study shows that summarizing video is time consuming and tedious, and that surgeons struggle to carve out time from their schedules to do this process, often delegating parts, even if the software they rely on does not support collaborative work, thus leading to ad-hoc coordination strategies. Although automatic methods based on image processing exist, they still lead to the inclusion of non-informative scenes, and require manual work to meet the expectations of surgeons [2].

Our long-term goal is to design tools that support surgeons in efficiently summarizing their videos. In this paper, we take a first step and study Avellino et al.'s [5] proposed approach of *tagging* segments to summarize surgical videos. Instead of the traditional *subtractive* approach of removing segments, this *additive* approach consists of selecting the relevant segments to keep through tags. In particular, we study:

- Q1: How does an additive approach, based on concatenating tagged video segments, support surgeons in summarizing surgical videos?
- Q2: How are tags used to support collaboration in the summarization process?

We conducted three workshops with two participants each, an expert ENT (Ears Nose Throat) surgeon and a resident. The pairs used a research prototype that lets them tag important steps and

key moments, edit them, and collaborate in real time. The results show that the additive approach can meet the needs of surgeons with certain limitations. Tagging was used to isolate interesting steps in surgery, although participants expressed the need for a subtractive approach during refinement work. Tags also proved useful for efficiently defining multiple summaries with different goals at the same time, from the same raw video. Regarding collaboration, we observe that tags can support coordination work, and that the resident's experience level affected how roles are distributed. These results inform the design of a subsequent controlled study, with the goal of comparing the subtractive approach with the traditional additive approach of commercial video editing software.

2 RELATED WORK

2.1 Practices Around Surgical Videos

Minimally-invasive surgery consists of inserting surgical instruments and a camera, called an endoscope, into the patient through small incisions. The surgery is then carried out by observing a video stream of the surgical site on a 2D screen. In this setup, video recording is straightforward as it only requires pressing a button to record [30]. However, the recorded video spans the entire duration of the surgery, in most cases more than an hour and sometimes up to ten hours or more. Therefore, for recorded videos to be usable, they need to be edited in order to include only relevant sequences, and usually reduced to less than twenty minutes [11]. This process is time consuming, resulting in recorded videos staying unedited, and therefore not used [5]. Once summarized, surgical videos can have a considerable impact, as they are an effective tool for pedagogy, especially in the training of junior surgeons, helping them visualize gestures and learn new procedures [6, 35]. They also contribute to the wide dissemination of surgical expertise in science.

2.2 Video Summarization

Video editing methods can be classified according to automation.

2.2.1 Manual editing. The manual method is the most used in the field of surgery, even if it is time consuming. Constructing a video summary involves multiple stages, where surgeons identify sequences of interest by iteratively removing surrounding sequences [5]. This constitutes a *subtractive* approach. Avellino et al. [5] posits that an *additive* approach, consisting of keeping the relevant parts, could be more efficient. Moreover, summarization is carried out collaboratively, where the expert surgeon delegates steps to junior surgeons or residents [3, 5], who may or may not have participated in the surgical procedure. This leads to several iterations between the pair for the final result to be achieved since the expert surgeon has a precise video summary in mind. In addition, as there is a considerable delay between the recorded surgery and the editing, sometimes several weeks, the surgeon spends time rewatching the original video to recall the timing of important moments [5]. The commercial editing tools used, such as iMovie (Apple), rely on the *subtractive* approach, requiring the surgeon to identify and remove irrelevant sequences, which often encompass the majority of the original video. Discarded sequences are difficult to recover, as they are deleted from the final edit. Moreover, these tools lack support for collaborative practices in video editing [5].

Finally, outsourcing video editing is not viable, as summarization requires in-depth knowledge of the field and thus surgeons do summaries themselves [3, 5, 12].

2.2.2 (Semi-)automatic editing. Automatic video processing methods rely on external or internal information to produce video summaries. Methods that rely on external information, such as audio [23], annotations or location data, are not relevant in surgery because the video does not contain these elements. Methods that rely on internal information, such as detecting gestures [22], colors [10] or movement activities [7] are also not adequate, because surgical videos are composed of a single shot with red toned frames, and thus do not include shot changes that are detectable (like in movies), or strong color variations. Certain systems aimed at surgery make it possible to detect unusable sequences, such as overexposed or blurred images [31], or images where the camera is occluded [25]. Some tools can also identify interesting sequences by recognizing relevant gestures or instruments in the video [26, 27]. Research tools, such as *Surch*, [19], or *EndoDigest* [29] can detect steps in a surgical video, which may ease video editing, although they have not been studied for summarization specifically. Other solutions let the user label a set of segments to train a system and let the system edit videos [9]. However, these systems have generally not been evaluated, except for *Magisto* [2], a system that creates video summaries using Artificial Intelligence (AI). In the evaluation, expert surgeons judged the resulting summaries as satisfactory only in 60% of cases.

All in all, with the exception of some specific contexts, e.g., surveillance camera video [8], automatic video summarization remains a challenge, especially in the surgical domain. This probably explains why these methods are not integrated into current practices and the editing process is left to the care of a surgeon.

2.3 Adding Semantics to Video Data

The addition of tags to videos has been explored in HCI. For example, Mackay [28] studied how researchers can tag video recordings of experiments, to navigate key moments during analysis. Huron et al. [15] explored the use of participative tags via Twitter to facilitate the analysis of scientific conference videos, while Perin et al. [34] investigated video annotation in the sports context, showing that spectators are capable of annotating key information accurately and in detail. Specifically in surgery, screenshots [24], and annotations combined with voice [12] have been proposed to mark video segments and ease frame retrieval. Although these work show that annotations make it possible to find desired segments, studies have not studied the usefulness of tags to create video summaries so far. Finally, a few works have explored tags and collaboration in the creation of video summaries. *VidCrit* [32] is designed so that someone who tags a video can interact with another person who gives feedback on the annotations. Despite the technical development of collaborative features, this work does not study how tags support collaborative work. *Polemictweet* [15] relies on the collaboration of scientific conference attendees, but the *tweets* are one-way, ultimately limiting the purely collaborative aspect. The *DIVER* project [33] proposes an interface for collaboratively annotating videos, associating tags with video segments. Although this enables for the creation of lists of labeled video segments, the

primary goal of this project is not video editing but conducting analytics on video through a more efficient navigation.

In summary, the most commonly used approach to video summarization remains manual editing, despite the various difficulties mentioned. Although video annotation has been studied, these studies do not focus on video summarization or collaboration.

3 METHOD

Our goal is to understand how the use of tags can contribute to the collaborative summarization of videos. We developed a prototype that implements an additive approach to summarization, as surgeons can tag video segments collaboratively, in real time, and a summary is defined as the concatenation of the tags. We chose expert workshops to study this prototype using pre-recorded videos, as the prototype implements a novel concept that cannot yet be used in a real setting—the operating room. For transparency, we have pre-registered the protocol and questionnaires: <https://osf.io/uh7bq>.

3.1 Participants

6 surgeons participated to 3 workshops in pairs. Each pair included one expert, an attending surgeon with more than 5 years post-residency experience, and one novice, a resident with varying level of experience (Table 1). Attending surgeons are senior doctors, who have finished their residency and post residency (post residency involves 2 to 4 years of further training and specialization after completing residency). A resident is a medical student, already specialized in a particular field, e.g., visceral surgery or cardiology, and working full time at the hospital. Participants were recruited through the authors’ personal network and through word of mouth. Participants were acquainted to different degrees: P1 and P2 had been working in the same service for 4 months, and P5 and P6 for one and a half years. The second workshop was the only one where the participants had not worked together previously.

Workshop	Participant	Role	Experience	Gender
W1	P1	Attending surgeon	Expert	Male
	P2	Resident	moderate	Female
W2	P3	Attending surgeon	Expert	Male
	P4	Resident	low	Female
W3	P5	Attending surgeon	Expert	Male
	P6	Resident	High	Female

Table 1: Participant demographics.

3.2 Apparatus

We name the prototype peroscope, for per-operative endoscopy (Figure 1). It is a web interface running on the research platforms Webstrates [20] and Videostrates [21]. The prototype lets users create tags on a video timeline as a mechanism to select footage to keep on a summary, and provides real-time collaborative functions to edit the tags. Users can first create tag categories, specifying a name and color, and then start the recording of a live video.

The video is displayed in the center left, the lower part shows two timelines, a full timeline which represents all of the recorded footage, and the summary timeline, computed as the concatenation

of all the tagged footage. The user can tag segments either as a “sequence”, tagging long segments, or, an “event”, with a predefined 5 second duration to tag short segments. Tags can be resized, renamed, moved, deleted, deactivated from the summary. These actions have an instant effect on the summary timeline. It is also possible to set a duration goal, which shades with background color the summary timeline. The recorded video can be navigated through hovering on both timelines, which seeks to the corresponding frame on the video footage. Multiple users can work on the same project, their changes are synchronized in-real time across instances.

3.3 Procedure

Before each workshop, we asked the attending surgeon to send us one or two videos of recorded surgeries, lasting about an hour. We received one video per attending surgeon, which we used to prepare an empty project. The day of the workshop, we welcomed the participants, explained the purpose of this study, and presented consent forms. We introduced the prototype, showing an example project to demonstrate the different functions, including how to create tag categories (define a name, change the color, deactivate), and how to work with tags (create in a specific position of the timeline, edit, reposition and resize). We also showed how to set a duration goal. The first and last authors led the workshop, which consisted of three parts.

3.3.1 Part I: Tagging to Summarize. The goal of this first part was to have participants tag the video during 10–15 minutes. We first asked participants to define several goals for the summary of the video they wanted to produce, for example, creating a course, presenting a technique at a conference, or a case for a surgery planning meeting. We selected one goal for the first task, and informed the pair of participants that this goal would be changed in the second part. The attending surgeon worked on the video alone while the resident remained an observer for this first part. We used a think-aloud protocol to understand the rationale behind actions.

3.3.2 Part II: Collaborative Work. For the second part, we originally planned to remove tags at random (less than 5%), operationalizing the fact that the operating room context may cause some important tags to be omitted during surgery. However, we did not carry out this deletion as participants created fewer tags than expected. We placed the participants on different computers, with the instruction to speak to each with other only if necessary when collaboratively editing the video. We aimed to observe communication through tags between the participants, for example, advice or instruction from the attending surgeon and questions or suggestions from the resident, as well as the division of the work between the pairs to avoid redundancies or conflicts. We also wanted to monitor latent needs, conflicts, or frustration points.

3.3.3 Part III: Repurposing of Tags. For the last part, we asked participants to change the goal of the video summary. For example, if the initial goal was to summarize for a conference (aimed at experts), we would ask them to now create a summary for producing learning material (aimed at junior surgeons). Our goal was to observe how the pair edits their tags, or appropriate different functions (names, colors, ability to deactivate) in order to reflect the goal modification.

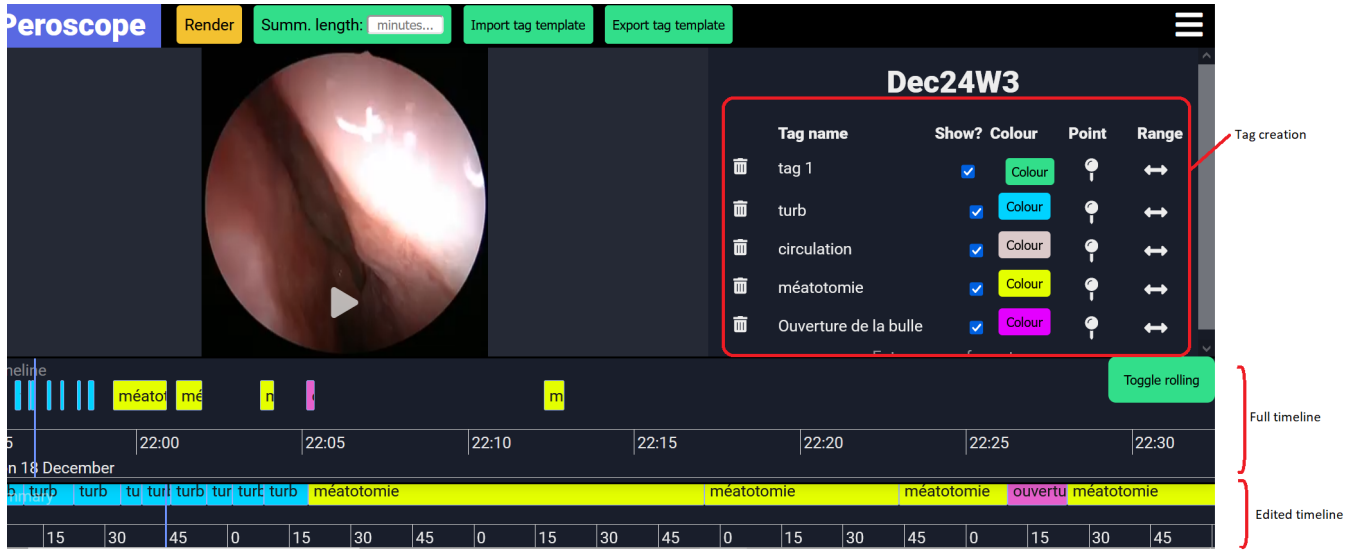


Figure 1: Prototype interface. The center left area shows a video with a play / pause button. On the center right area tag categories can be created and managed, as the user can define the name and color, delete the category, or deactivate the category from the summary timeline). Tags can be created, either as event or sequence. At the bottom of the screen, there are two timelines. The full timeline represents the complete recorded video, the user can seek to a location through hovering with the mouse. All the tags are present in this timeline, and users can move them or change their length by clicking on their edge and dragging the tag left or right. After selecting a tag, pressing the “s” key will split the tag at the location of the seek bar. A right click on a tag opens a menu with options to delete, change the name or the color, without modifying the others tags from the same tag category. The second timeline displays a concatenation of the active tags, and represents the final edit. The user can navigate the final summary by hovering with the mouse. At the top of the screen different options are accessible. The render button exports the edit as an MP4 file, and the user can define a duration goal for the video with the “summ. length” button, a colored background will appear on the edited timeline up to the duration goal.

After part III, we debriefed with the participants, to precise their impressions on the usefulness of tagging to summarize, the challenges they faced and the strategies they used to collaborate. We also harvested their ideas about how tags can be created in the operating room during surgery.

3.4 Data Collection and Analysis

The workshops were filmed using two cameras (one for each participant) and the first author took written notes. Participants completed a short questionnaire (available as supplementary material) after the workshop, focusing on the use of tags for summarization and for collaboration. To analyze the data, we first created a list of ways in which tags were used and appropriated, identifying strategies used in an additive approach too summarization, identifying challenges. We looked for patterns that reflect how different constraints affect these uses, for example, the fact that the objective changes, the relationship between tags and the content of the video, as well as collaborative practices through tags.

4 RESULTS

The study highlights the benefits and limitations of an additive approach through tags for summarization. Firstly, participants considered the prototype useful to easily produce multiple summaries from one original video, of different lengths and for different audiences. Second, we show that collaboration dynamics are influenced by the resident’s experience levels, as we observe a higher degree of synchronous collaboration with the senior surgeon when the resident is less experienced. Finally, the study emphasizes the need for more awareness of other user’s actions. We develop each of these results in this section.

4.1 Tags for Additive Summarization

Participants expressed that the concept can make summarization more efficient. For example, when we asked P6 about the use of this concept in daily practice, they answered “Yes, especially live in the OR”, as they can already start marking relevant segments while capturing video. P3 stated that tags would “allow to gain time on editing”. Concerning the editing process, we observed two main phases: the *coarse* editing and the *fine* editing, or refining stage. In the coarse editing phase, participants tag the main steps of the surgery, marking the beginning of each of these stages, and then extending the tag until the end of the phase. Then, they give this tag a name. We observe that by delimiting the surgical steps, a large part

of the useless sequences residing between these steps are automatically eliminated. In the subsequent phase, participants refined their tags by eliminating redundant gestures, such as repetitions in sutures, and long sequences without movement, such as cauterization gestures. While the first phase was completed by all participants, only two of the three senior participants completed the fine editing. They indicated that this is the most tedious part of editing, and does not always require experience in the field. Interestingly, we observed that participants wanted to also remove sequences inside already tagged sequences. This was done through editing existing tags, for example by trimming edges, or splitting them to delete middle segments, reducing their size. However, participants mentioned that the prototype did not facilitate fulfilling this need. The modification of the length of a tag, however, gave room for uncertainty about which parts should be included in the summary, as trimmed sequenced can easily be recovered. P1 expressed that *“It would be interesting to be able to erase part of tags”*, referring to a real physical eraser, mimicking with his hand an eraser being rubbed over parts of a long tag.

4.2 Tags for Creating Multiple Summaries

By the end of the first part, participants had created between 10 and 15 tags. All participants named the different tags, but at first, P1 and P5 did not use color to define categories, while P3 changed colors for each new surgical step they defined. We observe that these functions were used in different ways to create several summaries from the same full video. One participant used a category of tags to indicate sequences including all repetitive gestures, and another category to indicate only the first occurrence of the gesture. In this way, it was therefore possible to produce different summaries for different objectives by activating and deactivating categories of tags in the summary. For example, creating an educational summary for inexperienced residents by keeping all the repetitive gestures, and creating a conference video for senior surgeons by keeping only the first occurrence of the gesture. During the debriefing time, this notion appeared particularly interesting to all participants, as traditional editing software does not support this need. Finally, P1 suggested that the tags could be integrated into the video, to serve as subtitles or annotations.

4.3 Tags for Collaborative Work

We observed that the dynamics of the collaboration depends on the expertise of the resident. P2 and P4, who had less experience as residents, immediately asked for oral instructions, while P6, the most experienced resident, started fine editing without waiting for instructions. During the workshop with the least experienced resident (P4), the senior surgeon precisely defined the tasks to be carried out, since he asked him to fine edit a particular surgical step, after he showed him where this step was on the timeline through creating a tag. P4 also had more questions to ask the surgeon, as he struggled to differentiate two similar anatomical structures and to find the end of the surgical step. P2, more experienced, was instructed to edit the next steps of the surgery, he performed both coarse and fine sequencing, and did not need much advice, thus, oral communication remained lower than in the case of P4. P6, the most experienced resident, was quickly left alone, with P5 mainly

observing and giving very little advice. These observations show that an inexperienced resident generally have a harder time performing the coarse editing, because they encounter difficulties in identifying the main steps of the video and need guidance from the attending surgeon, whereas an experienced resident can perform the coarse editing as well as part of the fine editing, requiring only validation. For example, in the third workshop, P5 did not need to modify the edit, and simply checked the work of P6, which could have been done asynchronously. Conversely, in the second workshop, P4 was not able to continue working without the assistance of P3.

The use of tags to communicate appeared one time in W2, it was not the most used means of communication. The need to point to specific elements on the timeline appeared in all three cases. While the awareness of the working windows of other users was sufficient to avoid conflicts, it was insufficient for the resident or the surgeon to draw each other’s attention to a specific part on the timeline. For example, P3 said *“you have to work here”* while instinctively showing with his pointer the moment he was talking about on the timeline. Still, the pointer is not synchronized. This may indicate that synchronized tags may not fulfill all needs when communicating synchronously. Lastly, we observed two instances of using tags to delegate work. P1 and P3 asked their resident to work on a specific parts of the video, tagging that segment on the timeline.

4.4 Adequacy of Tagging in the Operating Room

The last result concerns the projections about the creation of tags in the operating room while performing surgery. While participants welcomed this idea, they expressed concerns given that surgical work remains the priority. P5 insisted on the importance of *“a simple interface”*, emphasizing that this can increase the likelihood of widespread utilization. P1 mentioned the importance of being aware that a tag is in progress during surgery, to avoid endless markings and omissions of markings with minimal distraction. However, it was admitted that errors and omissions were inevitable. All surgeons hypothesized that intraoperative tagging would remain rough, and should be corrected after the surgery. The difficulty of tagging a complication, e.g., an important bleeding, which induces stress and thus cognitive load was mentionned. As P1 suggested, it is simple to create a tag indicating that a crucial moment occurred earlier in the video. P5 stated that experienced surgeons could remain calm enough to tag a complication. The question of interaction to start and end a tag while performing surgery was systematically asked, without being explored because it was not the subject of our work.

5 DISCUSSION

While the use of tags has been studied in HCI, showing they can increase efficiency in retrieving interesting frames [28], our study suggests the relevance of tags for the creation of video summaries because they can serve, in addition to this purpose, as a video editing tool. Our findings show similar results to previous work regarding how summaries are created [5], and contribute with a first understanding of how tags can be used in support of video summaries.

5.1 Tags support Additive Summarization

5.1.1 Complement with the Subtractive Approach. Our study suggests that a single approach is not sufficient for creating video summaries in surgery. The additive approach seems effective for the preliminary steps, consisting in identifying relevant segments, while the subtractive approach seems more adequate during the refinement step to eliminate unnecessary sequences. We note that traditional video editing software make it difficult to retrieve a segment once it has been removed from the video edit, as it disappears from the timeline. Our prototype keeps the full timeline available for navigation, facilitating the identification of removed sequences.

5.1.2 Multiple Summaries. Commercial tools lack the flexibility to create multiple summaries from one video, leading to the repetition of work each time a different summary from the same video is needed [5]. The possibility of creating different tag types opened to the possibility of creating multiples edits from one video, the enthusiasm of the participants for this possibility underlines the notion that the pertinence of a summary varies with the targeted audience.

5.1.3 Integrating Automatic Tools. We also believe that the use of tags could be complementary with (semi-)automatic methods of video editing. For example, during *fine sequencing* steps, sequences with repetitive gestures or no movement can be tagged automatically by a tool such as *surch* [19], and blurry frames by *EndoDigest* [29]. On the other hand, tagging the main steps would remain at the discretion of the surgeon, as automatic methods, such as those based on Artificial Intelligence, are not yet reliable [2, 9].

5.2 Collaborative Work: Complement Between Decision and Workforce

We interpret senior surgeons delegating parts of the editing process to resident as a mean to save time, rather than to improve the quality of the final result by combining expertise, as suggested by previous work [3, 5]. Tags, even when employed as editing tools, maintain their function as markers to facilitate the retrieval of information within the videos. This added layer of information can facilitate summarization given how often the task is delegated and the time gap between surgery and editing. We add to this literature showing that, depending on their expertise, a different type and amount of tasks are delegated to the resident. Likewise, the amount of synchronous work depends on the expertise of the resident. Less-experienced residents benefit from working side by side with the senior, to receive help in recognizing anatomical structures and the different surgical steps in the video. The results therefore suggest that the need for synchronous work, is less necessary when the resident has more experience. The distribution of work, as well as the usefulness of synchronous work must be interpreted with caution because surgical videos are by nature often several hours long [5, 11]. Synchronous work can speed up the creation of surgical summaries by encouraging teamwork on long projects.

Finally, we emphasize, as mentioned by two thirds of the participants and the literature [3, 13], that the collaborative aspects of the concept can improve the education, given the need to reflect on content. By marking the steps and important gestures, the residents become aware of their level of understanding on a particular

surgery, and shows which concepts they are missing to achieve the final edited video. While the contribution of digital learning via virtual reality and educational videos is at the origin of multiple studies [16–18], our study paves the way to future work exploring the improvement of video editing through tags.

5.3 Limitations and Future Work

Our prototype had several limitations. It had minor bugs and significant buffering time, causing synchronization issues between the cursor position on the timeline and the corresponding video frame, which influenced the user experience. However, the fact that the participants still agreed on the validity of the concept reinforces the relevance of our results. Moreover, this is an exploratory study, and it does not compare the additive approach to regular editing tools in a controlled study. The next step is the introduction of the concept into the daily practice of surgeons, particularly in the operating room, which involves many obstacles such as taking into account the distraction specific to this environment [4], the cognitive load of the surgeon during surgery [14] and finding participants to test the concept in real-life practice. Finally, the main challenge will reside in evaluating the effectiveness of the concept by comparing the prototype to existing editing software in term of quality of the summaries produced, and the time needed to produce them.

6 CONCLUSION

We introduced and tested the concept of tagging to summarize collaboratively to surgeon-resident pairs. The concept was found relevant, useful and practical by participants in various ways. Several unexpected notions were highlighted, such as the relation between collaboration and level experience of the residents, the possibility of multiplying the summaries from a unique video through tags, and the pedagogic use enabled by collaboration and synchronous work. This work informs the design of future studies that can demonstrate the benefits of tagging to create video summaries through a controlled experiment that compares this approach with classic video editing tools.

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